

WAVE OPTICS

Wave Optics

Wave Nature of Light

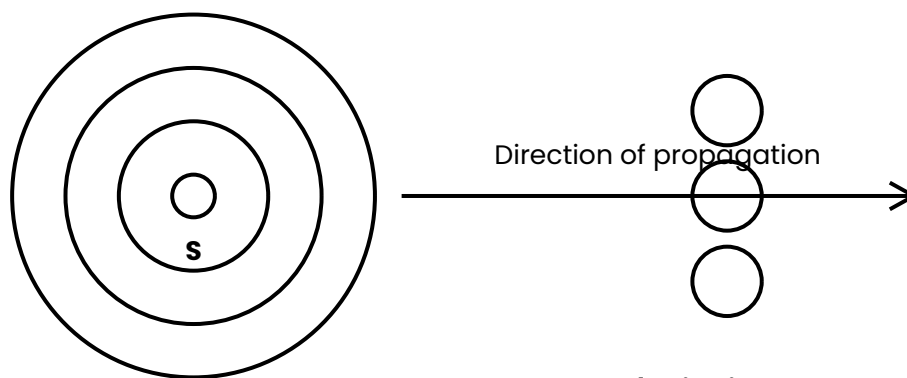
Wave optics explains the wave nature of light and the phenomena of interference, diffraction and polarisation. These phenomena cannot be explained completely by ray optics.

Wavefront

A wavefront is the locus of all points in a medium which are in the same phase of vibration at a given instant.

Types of Wavefront

- Spherical wavefront is produced by a point source of light.
- Plane wavefront is obtained when the source is very far away or when a small portion of a spherical wavefront is considered.
- The normal drawn to a wavefront at any point gives the direction of propagation of the wave.



Wavefront and Huygens' Principle

Huygens' Principle

According to Huygens' principle, every point on a wavefront acts as a source of secondary wavelets which spread in the forward direction with the speed of light in the medium.

- The new wavefront is the common tangent to the secondary wavelets.
- Huygens' principle explains the propagation of light and the laws of reflection and refraction.

Interference of Light

Interference is the redistribution of light intensity produced when two or more coherent light waves superpose.

Coherent Sources

- Coherent sources have the same frequency and maintain a constant phase difference.
- Coherent sources are required to obtain a steady interference pattern.

Constructive Interference

Constructive interference produces maximum intensity when the waves arrive in phase.

$$\Delta = n \lambda$$

$$\varphi = 2 n \pi$$

Destructive Interference

Destructive interference produces minimum intensity when the waves arrive out of phase.

$$\Delta = (2n + 1) \lambda / 2$$

$$\varphi = (2n + 1) \pi$$

Path Difference and Phase Difference

$$\varphi = 2 \pi \Delta / \lambda$$

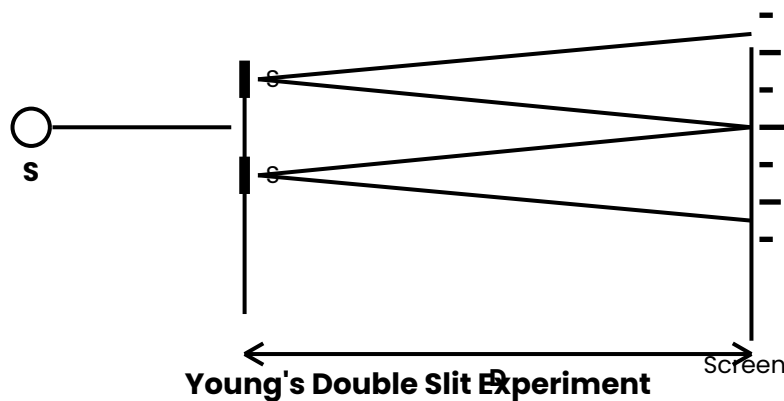
Intensity of Interference

$$I = I_1 + I_2 + 2 \sqrt{I_1 I_2} \cos(\varphi)$$

Young's Double Slit Experiment

Young's double slit experiment demonstrates the interference of light using two coherent sources obtained from a single source.

- The two narrow slits act as coherent sources.
- Alternate bright and dark fringes are observed on the screen.



Fringe Width

$$\beta = \lambda D / d$$

Bright Fringe

$$\Delta = n \lambda$$

Dark Fringe

$$\Delta = (2n + 1) \lambda / 2$$

Position of Bright Fringes

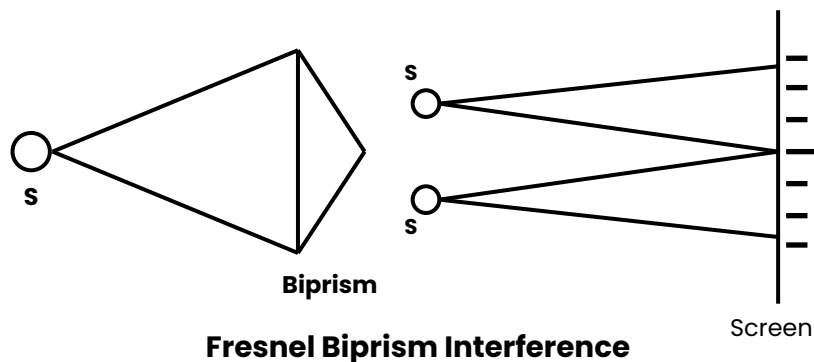
$$y_n = n \lambda D / d$$

Position of Dark Fringes

$$y_n = (2n + 1) \lambda D / 2d$$

Fresnel Biprism

A Fresnel biprism produces two virtual coherent sources from a single source. The interference pattern produced by these sources can be used to determine the wavelength of light.



$$\lambda = \beta d / D$$

Where:

- β = Fringe width

- d = Distance between coherent sources
- D = Distance between sources and screen

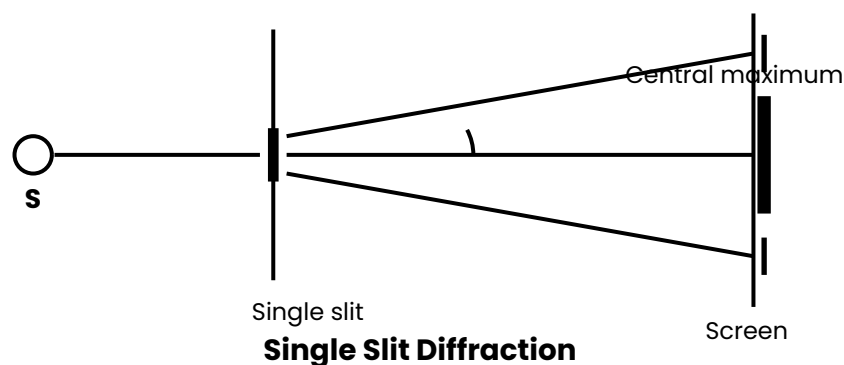
Diffraction of Light

Diffraction is the bending and spreading of light waves around the edges of an obstacle or through a narrow aperture.

- Diffraction becomes prominent when the size of the aperture or obstacle is comparable to the wavelength of light.

Single Slit Diffraction

In single slit diffraction, light passing through a narrow slit spreads into the region behind the slit and produces a diffraction pattern.



Condition for Minima

$$a \sin(\theta) = n \lambda$$

Width of Central Maximum

$$W = 2 \lambda D / a$$

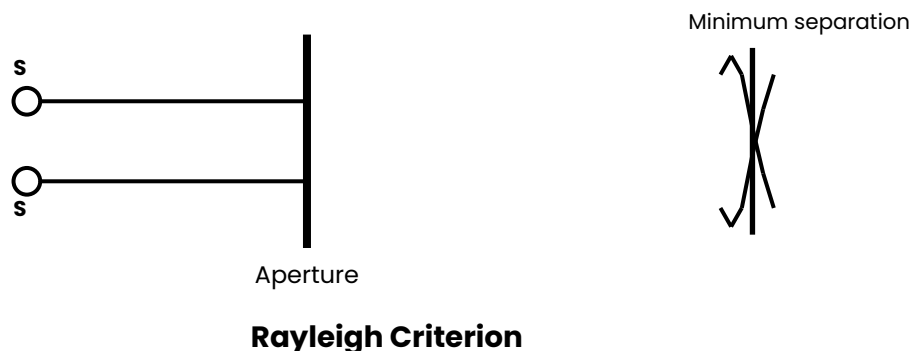
Interference and Diffraction

- Interference is due to superposition of waves from two or more coherent sources.
- Diffraction is the spreading of waves due to a narrow aperture or obstacle.
- Interference fringes are ideally of equal width in Young's double slit experiment.
- In single slit diffraction, the central maximum is wider than the secondary maxima.

Resolving Power and Rayleigh Criterion

The resolving power of an optical instrument indicates its ability to distinguish two closely spaced objects as separate.

According to Rayleigh's criterion, two point objects are just resolved when the principal maximum of one diffraction pattern coincides with the first minimum of the other.



Resolving Power of Telescope

$$\theta_{\min} = 1.22 \lambda / D$$

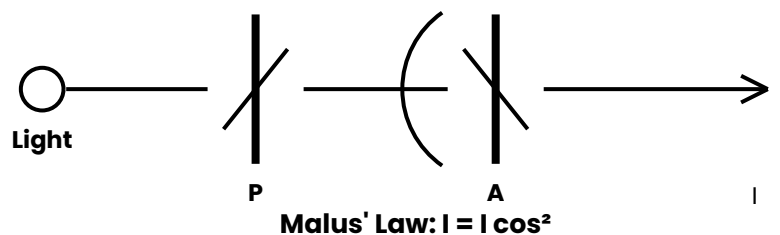
Polarisation of Light

Polarisation is the phenomenon of restricting the vibrations of light waves to a particular direction perpendicular to the direction of propagation.

- Only transverse waves can be polarised.
- Polarisation proves the transverse nature of light.

Polaroids

A polaroid is a material that transmits light vibrations mainly along one particular direction and absorbs vibrations in the perpendicular direction.



Polarisation and Malus' Law

Malus' Law

$$I = I_0 \cos^2(\theta)$$

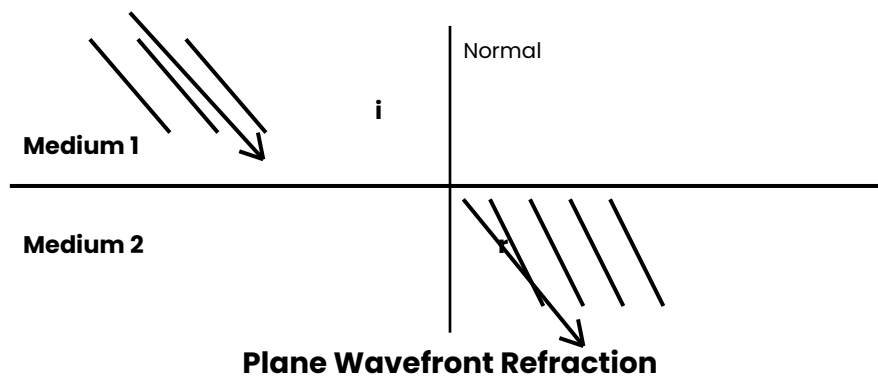
Brewster's Law

$$\mu = \tan(i_p)$$

- At the polarising angle, the reflected and refracted rays are mutually perpendicular.

Reflection and Refraction Using Wavefronts

Huygens' principle can be used to explain reflection and refraction of a plane wavefront at a surface separating two media.



Doppler Effect in Light

The Doppler effect in light is the apparent change in frequency or wavelength of light due to relative motion between the source and the observer.

Quick Revision Points

- Wavefront is the locus of points having the same phase.
- Huygens' principle explains the propagation of wavefronts.
- Coherent sources have the same frequency and constant phase difference.
- Constructive interference gives maximum intensity.
- Destructive interference gives minimum intensity.
- Fringe width in YDSE is proportional to wavelength and screen distance.
- Fringe width is inversely proportional to slit separation.

- Diffraction becomes significant when aperture size is comparable to wavelength.
- The central maximum in single slit diffraction is twice the width of the secondary maxima.
- Rayleigh criterion gives the condition for just resolving two point objects.
- Polarisation establishes the transverse nature of light.
- Malus' law gives the intensity transmitted through an analyzer.

Important Formulae

$$\varphi = 2 \pi \Delta / \lambda$$

$$I = I_1 + I_2 + 2 \sqrt{I_1 I_2} \cos(\varphi)$$

$$\beta = \lambda D / d$$

$$y_n = n \lambda D / d$$

$$y_n = (2n + 1) \lambda D / 2d$$

$$\lambda = \beta d / D$$

$$a \sin(\theta) = n \lambda$$

$$W = 2 \lambda D / a$$

$$\theta_{\min} = 1.22 \lambda / D$$

$$I = I_0 \cos^2(\theta)$$

$$\mu = \tan(i_p)$$

Conclusion

Wave optics explains the wave nature of light through wavefronts, Huygens' principle, interference, diffraction and polarisation. These concepts are important for understanding the behaviour of light and solving numerical problems.